TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	79.1 ± 2.77
Correlation Stability	87.8 ± 2.23
Distribution Consistency	86.4 ± 2.54
Entropy Preservation	88.5 ± 1.43
Mean Z Score Drift	0.2 ± 0.03
Categorical Fidelity	82.8 ± 0.40
Privacy Risk	49.3 ± 1.27
Data Density	99.9 ± 0.14
Coverage Score	22.3 ± 4.41
Feature Accuracy	78.3 ± 2.24
Novelty Score	0.5 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	42.079	-2.667	6.0%	OK
TB	3.299	1.756	-1.543	46.8%	Drift
DB	1.486	0.752	-0.735	49.4%	Drift
Alkphos	290.576	223.487	-67.089	23.1%	Warn
Sgpt	80.714	38.144	-42.570	52.7%	Drift
Sgot	109.911	39.719	-70.192	63.9%	Drift
TP	6.483	6.826	+0.343	5.3%	OK
ALB	3.142	3.508	+0.366	11.7%	Warn
A/G Ratio	0.947	1.049	+0.102	10.8%	Warn
Selector	1.286	1.438	+0.151	11.8%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	79.1 ± 2.77
Correlation Stability (*2)	87.8 ± 2.23
Distribution Consistency (*3)	86.4 ± 2.54
Entropy Preservation (*4)	88.5 ± 1.43
Mean Z Score Drift (*5)	0.2 ± 0.03
Categorical Fidelity (*6)	82.8 ± 0.40
Privacy Risk (*7)	49.3 ± 1.27
Data Density (*8)	99.9 ± 0.14
Coverage Score (*9)	22.3 ± 4.41

Feature Accuracy (*10)	78.3 ± 2.24
Novelty Score (*11)	0.5 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.5452991452991454	0.7145299145299144	0.011111111111111117	
F1	0.8108734553482932	0.6574105617775727	0.8198043337473587	0.008930878399065456	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6846153846153846	0.5410256410256411	0.6940170940170940	0.00940170940170939	
F1	0.7903670764722801	0.6358960807773331	0.7965013103440624	0.006134233871782235	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6598290598290599	0.5188034188034187	0.6401709401709402	-0.019658119658119633	
F1	0.7629912434171616	0.609939204334832	0.7504475811779372	-0.012543662239224406	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.5726495726495726	0.6931623931623931	0.018803418803418813	
F1	0.7821343746683586	0.6775322054046636	0.8028045591724229	0.020670184504064402	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6196581196581197	0.670940170940171	-0.015384615384615396	
F1	0.7971299319768493	0.7351412288277026	0.7855924084590332	-0.011537523517816162	

Model: XGB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.5598290598290598	0.6905982905982906	-0.016239316239316248	
F1	0.8163860452461005	0.6710008680463849	0.8059432085741476	-0.010442836671952982	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376069	0.5564102564102564	0.7059829059829059	-0.0008547008547008627	
F1	0.8096642085470265	0.6647158634163863	0.8126639762240696	0.002999767677043302	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6777777777777778	0.7068376068376068	0	
F1	0.8279091624529537	0.7923735875397167	0.8279091624529537	0	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.5829059829059828	0.5658119658119658	0.006837606837606836	
F1	0.5587432119380917	0.6873328134297972	0.5735367074173324	0.014793495479240793	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	73	Male	1.9	0.7	1750	102	141	5.5	2.0	0.5
2	60	Male	2.2	1.0	271	45	52	6.1	2.9	0.9
3	40	Male	3.6	1.8	285	50	60	7.0	2.9	0.7
4	45	Female	0.7	0.2	153	41	42	4.5	2.2	0.9
5	62	Male	5.0	2.1	103	18	40	5.0	2.1	1.72
6	50	Male	0.7	0.2	206	18	17	8.4	4.2	1.0
7	34	Female	1.2	0.4	210	30	25	6.5	3.0	1.1
8	45	Male	2.5	1.1	300	50	60	7.0	3.5	1.5
9	29	Female	0.8	0.2	150	20	18	5.8	2.9	0.9
10	52	Male	3.0	1.5	400	70	80	7.5	3.8	1.2
11	67	Female	1.5	0.5	220	45	50	6.2	3.1	0.8
12	39	Male	0.9	0.3	175	25	22	6.7	3.4	1.0
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1
2	75	Male	1.4	0.4	215	50	30	5.9	2.6	0.7
3	28	Female	0.9	0.2	316	25	23	8.5	5.5	1.8
4	49	Male	2.0	0.6	209	48	32	5.7	3.0	1.1
5	48	Male	0.7	0.2	326	29	17	8.7	5.5	1.7
6	58	Female	0.8	0.2	130	24	25	7.0	4.0	1.3
7	29	Female	1.2	0.4	210	28	30	6.5	3.4	1.2
8	54	Male	2.5	1.0	300	45	50	7.2	3.8	0.9
9	37	Female	0.8	0.2	175	22	24	6.0	3.1	1.0
10	62	Male	3.1	1.5	250	60	70	8.0	4.2	0.8
11	41	Female	0.9	0.3	190	30	28	6.8	3.5	1.1
12	50	Male	1.5	0.7	220	40	45	7.5	3.9	0.9
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	17	Female	0.5	0.1	206	28	21	7.1	4.5	1.7
2	26	Male	1.0	0.3	163	48	71	7.1	3.7	1.0
3	47	Male	0.9	0.2	265	40	28	8.0	4.0	1.0
4	50	Male	1.7	0.8	331	36	53	7.3	3.4	0.9
5	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
6	50	Male	0.7	0.2	192	18	15	7.4	4.2	1.3
7	25	Female	0.6	0.2	150	22	20	7.0	4.0	1.5
8	34	Male	1.2	0.4	210	35	30	6.5	3.5	1.0
9	42	Female	0.9	0.3	180	28	25	7.2	4.2	1.2
10	29	Male	1.5	0.5	220	40	35	6.8	3.8	1.1
11	53	Female	0.7	0.2	160	30	28	7.1	4.1	1.3
12	38	Male	1.0	0.3	190	33	32	6.9	3.9	1.0
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	13	Female	0.7	0.1	182	24	19	8.9	4.9	1.2
2	60	Male	11.0	4.9	750	140	350	5.5	2.1	0.6
3	26	Male	1.0	0.3	163	48	71	7.1	3.7	1.0
4	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62
5	50	Male	2.6	1.2	415	407	576	6.4	3.2	1.0
6	58	Male	1.0	0.5	158	37	43	7.2	3.6	1.0
7	25	Female	1.2	0.4	200	30	40	6.5	3.5	0.9
8	34	Male	0.8	0.2	180	25	35	7.0	3.8	1.1
9	45	Female	2.5	1.0	300	50	60	7.5	4.0	0.8
10	52	Male	1.0	0.3	220	40	45	6.8	3.2	1.0
11	29	Female	0.9	0.2	210	20	30	6.2	3.0	1.1
12	38	Male	3.0	1.5	400	60	70	8.0	4.5	0.7
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Female	0.8	0.2	165	22	18	8.2	4.1	1.0
2	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
3	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
4	42	Male	6.8	3.2	630	25	47	6.1	2.3	0.6
5	48	Female	1.4	0.8	621	110	176	7.2	3.9	1.1
6	62	Male	1.2	0.4	195	38	54	6.3	3.8	1.5
7	34	Female	1.2	0.4	210	25	30	6.5	3.2	1.1
8	29	Male	0.9	0.3	150	20	25	6.0	3.0	1.0
9	52	Female	2.5	1.0	300	45	50	7.0	3.5	0.9
10	41	Male	3.1	1.5	400	60	70	7.5	3.8	1.2
11	38	Female	0.8	0.2	180	15	20	5.5	2.8	0.8
12	45	Male	1.5	0.6	250	30	35	6.8	3.1	1.0
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Female	0.6	0.1	185	17	26	6.6	2.9	0.7
2	22	Male	2.7	1.0	160	82	127	5.5	3.1	1.2
3	66	Male	1.1	0.5	167	13	56	7.1	4.1	1.36
4	60	Male	2.1	1.0	191	114	247	4.0	1.6	0.6
5	62	Male	1.8	0.9	224	69	155	8.6	4.0	0.8
6	18	Male	0.7	0.1	312	308	405	6.9	3.7	1.1
7	34	Female	1.2	0.4	210	45	50	6.5	3.1	0.85
8	29	Male	0.8	0.3	180	30	25	5.9	2.8	0.93
9	55	Female	2.5	1.1	300	60	70	7.2	3.9	1.0
10	40	Male	1.0	0.5	220	50	40	6.0	2.5	0.75
11	22	Female	0.6	0.2	150	20	15	5.5	2.1	0.95
12	38	Male	3.0	1.5	250	70	80	7.0	3.5	0.7
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Male	1.5	0.4	298	60	103	6.0	3.0	1.0
2	60	Male	0.9	0.3	168	16	24	6.7	3.0	0.8
3	49	Female	0.8	0.2	198	23	20	7.0	4.3	1.5
4	65	Male	7.9	4.3	282	50	72	6.0	3.0	1.0
5	42	Male	11.1	6.1	214	60	186	6.9	2.8	2.8
6	70	Male	1.7	0.5	400	56	44	5.7	3.1	1.1
7	29	Female	0.7	0.2	150	20	25	6.5	3.5	1.0
8	34	Male	1.2	0.4	180	30	40	7.0	4.0	1.2
9	45	Female	2.5	1.0	220	40	50	6.8	3.2	0.8
10	50	Male	3.0	1.5	250	50	60	7.5	3.0	0.6
11	38	Female	0.9	0.3	200	25	30	6.2	3.8	1.1
12	62	Male	5.0	2.5	300	70	80	7.2	2.9	0.4
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	8.9	4.5	272	31	61	5.8	2.0	0.5
2	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
3	37	Male	1.3	0.4	195	41	38	5.3	2.1	0.6
4	16	Male	0.7	0.2	418	28	35	7.2	4.1	1.3
5	6	Male	0.6	0.1	289	38	30	4.8	2.0	0.7
6	38	Male	0.8	0.2	208	25	50	7.1	3.7	1.0
7	29	Female	1.2	0.5	210	35	50	6.5	3.2	0.8
8	45	Male	0.9	0.3	150	40	60	7.0	3.5	0.9
9	38	Female	2.0	0.9	300	45	70	6.8	3.6	1.0
10	52	Male	1.5	0.7	400	50	80	7.2	3.8	0.95
11	33	Female	0.8	0.2	180	30	40	6.0	3.0	0.75
12	60	Male	3.1	1.5	500	60	90	7.5	4.0	1.1
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Male	2.2	0.8	209	25	20	8.0	4.0	1.0
2	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
3	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
4	46	Male	0.6	0.2	115	14	11	6.9	3.4	0.9
5	17	Male	0.9	0.2	279	40	46	7.3	4.0	1.2
6	45	Female	0.9	0.3	189	23	33	6.6	3.9	nan
7	34	Female	1.2	0.4	210	30	25	6.5	3.5	1.1
8	29	Male	0.8	0.3	180	20	15	7.0	4.0	1.2
9	47	Female	2.5	1.1	300	40	35	6.8	3.9	1.0
10	55	Male	0.9	0.2	250	50	45	7.2	3.8	1.3
11	38	Female	1.5	0.5	190	28	22	6.0	3.2	1.1
12	60	Male	0.7	0.2	230	35	30	6.4	3.1	1.0
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	0.8	0.2	135	30	29	7.2	4.4	1.5
2	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
3	50	Male	2.6	1.2	415	407	576	6.4	3.2	1.0
4	65	Male	1.4	0.6	260	28	24	5.2	2.2	0.7
5	45	Male	2.2	0.8	209	25	20	8.0	4.0	1.0
6	55	Male	0.9	0.2	190	25	28	5.9	2.7	0.8
7	29	Female	1.2	0.4	180	22	30	6.5	3.2	1.1
8	34	Male	0.8	0.3	150	28	25	7.0	4.0	1.5
9	42	Female	2.5	1.0	200	35	40	6.8	3.5	1.2
10	55	Male	3.1	1.5	250	45	50	7.5	4.2	1.3
11	60	Female	1.0	0.5	170	30	20	8.0	3.8	1.0
12	48	Male	2.3	0.9	220	40	45	6.2	3.0	1.1
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3